Buckling Analysis of Cylindrical Shells using Bogner-Fox-Schmit (BFS) Elements

Siddharthan Somasundaram (Matriculation No.: 254304)[a,1]

[a]*Otto-von-Guericke-Universität Magdeburg, Germany*

Supervisor: Dr.-Ing. Sascha Eisenträger Institute of Materials, Technologies and Mechanics — Computational Mechanics Prof. Dr.-Ing. Daniel Juhre

Abstract—This project proposes the implementation and validation of a linear buckling analysis framework for cylindrical shells using the classical Bogner-Fox-Schmit (BFS) finite element. BFS elements are C^1-continuous plate and shell elements constructed from bi-cubic Hermitian polynomials, making them well-suited for thin-shell stability problems. Building upon an existing validated Matlab code base for BFS-based flat plate analysis, this work extends the formulation to cylindrical shell geometry through three structured phases: the derivation of the cylindrical shell element with 48 degrees of freedom per element; the implementation of a pre-buckling in-plane stress solver; and the assembly of the geometric stiffness matrix required for eigenvalue-based buckling analysis. The project targets a simply-supported cylindrical shell under uniform axial compression and validates the computed critical buckling load against the classical Donnell analytical solution and Ansys reference results. This work constitutes the direct computational foundation for the subsequent Master's thesis, which will extend the methodology to cylindrical shells with multiple cutouts using the Cut-BFS (CutBFS) approach based on the Finite Cell Method.

Keywords—*Bogner-Fox-Schmit elements, cylindrical shell, buckling analysis, geometric stiffness, Finite Cell Method, CutBFS*

Contents

1 Introductory Paragraph 1
- 1.1 Problem Statement and Importance . . . 1
- 1.2 Scope of the Project . . . 1

2 Brief State of the Art 1
- 2.1 Review of Relevant Literature . . . 1
- 2.2 Identified Gap . . . 1

3 Open Research Question 2

4 Methodology 2
- 4.1 Objectives and Techniques . . . 2
  - 4.1.0.1 Phase 1 — Cylinder geometry and mesh. 2
  - 4.1.0.2 Phase 2 — Pre-buckling in-plane stress analysis. . . . 2
  - 4.1.0.3 Phase 3 — Geometric stiffness and buckling eigenvalue. . . . 2
  - 4.1.0.4 Phase 4 — Validation and post-processing. 2
- 4.2 Simulation Tools . . . 2
- 4.3 Anticipated Limitations . . . 2

5 Outcome 2
- 5.1 Planned Results . . . 2
- 5.2 Contribution to Knowledge . . . 2

6 Conclusion 2

1. Introductory Paragraph

1.1. Problem Statement and Importance

Thin-walled cylindrical shell structures are fundamental load-bearing components in modern engineering applications, including aerospace fuselages, rocket bodies, submarine pressure hulls and offshore pipelines. Under axial compressive loading, these structures are susceptible to a sudden and catastrophic loss of stability known as *buckling* — a failure mode that is particularly dangerous because it occurs abruptly, often at load levels well below the material yield strength. The accurate prediction of the critical buckling load and the associated deformation mode shape is therefore an essential requirement in the structural design and certification of such components **timoshenko1961**, **abaqus2009**.

The Bogner-Fox-Schmit (BFS) element, introduced in 1965, is a high-efficiency C^1-continuous finite element based on a bi-cubic Hermitian polynomial approximation **bogner1965**. This continuity property makes it particularly well-suited for the analysis of thin shells under bending and buckling, as it analytically satisfies slope continuity between adjacent elements without penalty or Lagrange multiplier techniques. Despite its proven accuracy and implementation simplicity, the BFS element has seen limited application in modern structural analysis, primarily because its requirement for a structured rectangular mesh restricts its direct use on geometrically complex structures. Extending the BFS element to cylindrical shell geometries represents a natural and important step towards broader application in engineering practice **eisentraeger2022**.

1.2. Scope of the Project

This project implements and validates a linear buckling analysis framework for cylindrical shells using BFS elements. The work builds directly upon an existing validated Matlab code base for BFS-based plate analysis **eisentraeger2022** and extends it to curved shell geometry through three structured phases:

1. Formulation and implementation of the cylindrical BFS shell element with 48 degrees of freedom per element (12 per node, accounting for three displacement components);
2. Development of the pre-buckling in-plane (membrane) stress solver;
3. Construction and assembly of the geometric stiffness matrix $\mathbf{K}_G$ required for eigenvalue-based buckling analysis.

The project considers a simply-supported cylindrical shell under uniform axial compression — the canonical benchmark for shell buckling — and validates results against the classical Donnell analytical solution. This work constitutes the direct computational foundation for the subsequent Master's thesis, which will extend the methodology to cylindrical shells with multiple cutouts using the CutBFS approach.

2. Brief State of the Art

2.1. Review of Relevant Literature

Linear buckling analysis of cylindrical shells has a long history in structural mechanics. The classical analytical solution for a simply-supported cylindrical shell under axial compression, derived from Donnell's shallow shell equations, provides the reference critical stress **timoshenko1961**:

$$\sigma_{cr} = \frac{E}{\sqrt{3\,(1-\nu^2)}} \cdot \frac{h}{R}, \tag{1}$$

where E is Young's modulus, ν is Poisson's ratio, h is the wall thickness, and R is the cylinder radius. This expression remains the primary benchmark for numerical methods. Finite element analysis of shell buckling has advanced significantly, with commercial codes such as Abaqus and Ansys offering robust shell element formulations validated against experimental data **abaqus2009**.

The original BFS element formulation for cylindrical shells was proposed by Bogner, Fox and Schmit themselves **bogner1965**, demonstrating the element's suitability for curved geometries. More recently, Eisenträger et al. **eisentraeger2022** demonstrated that BFS elements, when combined with the Finite Cell Method (FCM) in the so-called CutBFS approach, can accurately predict the critical buckling load of rectangular plates with multiple cutouts. The CutBFS method replaces the geometry-conforming mesh requirement with a structured Cartesian background mesh, thereby retaining the accuracy and simplicity of BFS elements while enabling the analysis of geometrically complex structures. In a complementary development, Burman et al. **burman2020** established the mathematical foundation for cut BFS elements and proved optimal convergence rates in the context of plate bending. Furthermore, the Shell Finite Cell Method **shellFCM2011** demonstrated that fictitious domain approaches can be generalised to thin-walled shell structures using high-order p-version elements; however, no equivalent framework based on the C^1-continuous BFS element has been developed for cylindrical shell geometries.

2.2. Identified Gap

The central gap in the existing literature is the absence of a validated, open-source BFS-based buckling analysis framework for cylindrical shells. While BFS plate buckling has been comprehensively studied **eisentraeger2022**, and while cylindrical shell buckling has been analysed using alternative finite element formulations **timoshenko1961**, **abaqus2009**, the combination of BFS elements with cylindrical shell geometry and buckling analysis has not been implemented or validated in a modern Matlab environment. This project directly addresses this gap and provides the validated shell code base required for the CutBFS cylinder extension in the Master's thesis.

3. Open Research Question

The central research question of this project is:

> *Can BFS elements, implemented within the existing CutBFS Matlab framework, accurately predict the critical buckling load and mode shape of a simply-supported cylindrical shell under axial compression, and be validated against the classical Donnell analytical solution?*

This overarching question is addressed through three specific and graduated sub-questions:

1. **Pre-buckling stress:** What is the in-plane membrane stress distribution in the cylindrical shell wall under a unit axial compressive load, and does the BFS-based plane stress solver reproduce the expected uniform stress state?
2. **Critical load:** What is the critical load factor λ predicted by the BFS eigenvalue analysis, and how does it compare to the Donnell analytical solution given in Equation (1)?
3. **Mode shape:** What is the first buckling mode shape (number of axial half-waves m and circumferential waves n), and does it match the analytically predicted wave pattern?

4. Methodology

4.1. Objectives and Techniques

The project is structured into four implementation phases, each building directly on the previous:

Phase 1 — Cylinder geometry and mesh. The cylindrical shell of radius R, length L, and wall thickness h is *unfolded* into a rectangular domain using the coordinate transformation x (axial direction) and $y = R\theta$ (circumferential direction). A structured rectangular mesh is generated, enabling direct reuse of the existing BFS element infrastructure. The 48-DOF per element formulation is adopted, with each of the three displacement components (u, v, w) carrying four Hermitian DOFs per node:

$$f_i, \quad \left(\frac{\partial f}{\partial x}\right)_i, \quad \left(\frac{\partial f}{\partial y}\right)_i, \quad \left(\frac{\partial^2 f}{\partial x\,\partial y}\right)_i, \quad f \in \{u,\, v,\, w\}. \tag{2}$$

Phase 2 — Pre-buckling in-plane stress analysis. A 32-DOF plane stress stiffness matrix $\mathbf{K}_{\text{plane}}$ is assembled using the BFS shape functions for the in-plane displacement components u and v. A unit axial compressive load is applied as energetically equivalent nodal forces at the cylinder ends. The resulting membrane stress resultants ($\tilde{N}_x$, $\tilde{N}_y$, $\tilde{N}_{xy}$) are computed at each integration point and stored for Phase 3:

$$\mathbf{K}_{\text{plane}}\,\mathbf{U}_{\text{ref}} = \mathbf{F}_{\text{edge}}. \tag{3}$$

Phase 3 — Geometric stiffness and buckling eigenvalue. The geometric (initial-stress) stiffness matrix $\mathbf{K}_G$ is assembled by integrating the product of the BFS shape function gradients and the pre-buckling membrane stress tensor over each element:

$$\mathbf{k}_G^{(e)} = \int_{-1}^{1}\int_{-1}^{1} \mathbf{L}_g^{\top}\,\boldsymbol{\Sigma}\,\mathbf{L}_g\;\det(\mathbf{J})\,\mathrm{d}\xi\,\mathrm{d}\eta, \tag{4}$$

where $\mathbf{L}_g$ is the gradient operator applied to the BFS shape functions and $\boldsymbol{\Sigma}$ is the pre-buckling stress matrix. The generalised eigenvalue problem

$$\left(\mathbf{K}_{\text{bending}} + \lambda\,\mathbf{K}_G\right)\boldsymbol{\Phi} = \mathbf{0} \tag{5}$$

is solved using Matlab's `eigs` function to extract the critical load factor λ and the corresponding mode shape $\boldsymbol{\Phi}$.

Phase 4 — Validation and post-processing. The computed critical load is compared against the Donnell analytical solution (Eq. (1)). Convergence with respect to mesh refinement is assessed. The buckling mode shape is visualised using Matlab's `surf` command and compared qualitatively against the analytically predicted diamond patterns.

4.2. Simulation Tools

All implementation is carried out in Matlab, extending the existing BFS plate code provided by the research group **eisentraeger2022**. No commercial finite element software is used for the primary analysis; Ansys Mechanical is employed solely as an independent reference for cross-validation of the critical buckling load.

4.3. Anticipated Limitations

The analysis is restricted to *linear* (classical) buckling, which neglects geometric nonlinearities and initial imperfections known to significantly reduce the actual buckling load of cylindrical shells in practice. The simply-supported boundary condition is assumed throughout; clamped or free-edge conditions are not considered in this project but are identified as natural extensions. Cutouts are explicitly excluded from this project scope and are reserved for the Master's thesis.

5. Outcome

5.1. Planned Results

The expected technical outcomes of this project are:

- A validated Matlab implementation of the 48-DOF cylindrical BFS shell element, including the pre-buckling plane stress solver and the geometric stiffness matrix $\mathbf{K}_G$.
- A convergence study demonstrating that the BFS-based critical buckling load converges to the Donnell analytical solution with mesh refinement, with a target accuracy of less than 5%.
- Contour plots of the pre-buckling membrane stress distribution, confirming the expected uniform axial stress state under end compression.

- Visualisation of the first buckling mode shape on the cylindrical surface, showing the characteristic diamond pattern with the analytically predicted wave numbers m and n.
- A parametric study of the effect of the radius-to-thickness ratio R/h on the critical buckling load, comparing BFS results against the analytical formula and Ansys reference solutions.

5.2. Contribution to Knowledge

This project provides the first validated implementation of BFS shell elements for cylindrical shell buckling analysis within the existing CutBFS Matlab framework. It establishes a tested and documented code base that directly enables the Master's thesis extension to cylindrical shells with multiple cutouts using the CutBFS method. The parametric validation against analytical and commercial FEM solutions provides confidence in the correctness of the implementation and quantifies the accuracy of the BFS element for this class of problem.

6. Conclusion

This project proposes the implementation and validation of a linear buckling analysis framework for cylindrical shells using Bogner-Fox-Schmit elements, extending an existing validated Matlab code base for BFS plate analysis. The work is strategically scoped as the direct computational foundation for the subsequent Master's thesis on CutBFS-based buckling analysis of cylindrical shells with multiple cutouts.

By completing this project, the key missing components — the 48-DOF cylindrical shell element, the pre-buckling plane stress solver, and the geometric stiffness matrix $\mathbf{K}_G$ — will be implemented and validated, leaving the thesis to focus on the novel CutBFS extension with cutout handling and parametric optimisation. The immediate next step is a detailed study of the existing plate code structure alongside the mathematical roadmap document provided by the supervisor, before implementation begins in Matlab.

References

[1] S. Timoshenko, J. Gere, *Theory of Elastic Stability*, 2nd ed. McGraw-Hill, New York, 1961.

[2] Dassault Systèmes Simulia Corp., *Abaqus Theory Manual*, vol. 6.6, sec. 3.6.4. Providence, RI, USA, 2009.

[3] F. K. Bogner, R. L. Fox, L. A. Schmit, "The generation of interelement compatible stiffness and mass matrices by the use of interpolation formulae," in *Proc. 1st Conf. Matrix Methods in Structural Mechanics*, Wright Patterson AF Base, Ohio, 1965, pp. 397–444.

[4] S. Eisenträger, J. Kiendl, G. Michaloudis, R. Duy, Y. Vetyukov, "Stability analysis of plates using cut Bogner-Fox-Schmit elements," *Computers & Structures*, vol. 270, p. 106854, 2022. doi:10.1016/j.compstruc.2022.106854

[5] E. Burman, P. Hansbo, M. G. Larson, "Cut Bogner-Fox-Schmit elements for plates," *Adv. Model. Simul. Eng. Sci.*, vol. 7, no. 27, 2020. doi:10.1186/s40323-020-00164-3

[6] D. Schillinger, M. Ruess, N. Zander, Y. Bazilevs, A. Düster, E. Rank, "Small and large deformation analysis with the p- and B-spline versions of the Finite Cell Method," *Comput. Methods Appl. Mech. Eng.*, vol. 211, 2011.

[7] R. Duy, *Fictitious Domain Approach for Optimizing Stability Boundaries of Plates with Cutouts*, Master's thesis, TU Wien, Vienna, Austria, 2021.